

Post-Quantum Cryptography for Critical Infrastructure in Saudi Arabia and the GCC

Implementation Challenges, Reproducible Benchmarks
and a Migration Roadmap

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Abstract

The economies of Saudi Arabia and the other Gulf Cooperation Council (GCC) states depend on critical infrastructure (CI) that is dominated by operational technology: oil and gas production, petrochemicals, electric power and the world’s largest desalination fleet. The region has also experienced some of the most consequential cyber attacks on industrial systems, including Shamoon at Saudi Aramco in 2012 and the Triton attack on safety systems at a Saudi petrochemical facility in 2017. Cryptographically relevant quantum computers would break the RSA and elliptic-curve cryptography that authenticates and protects these systems today. Saudi Arabia’s National Cybersecurity Authority (NCA) has addressed post-quantum cryptography (PQC) in its National Cryptographic Standards since 2020 and consulted in December 2025 on an update that adds post-quantum algorithm standards, while the United Arab Emirates approved a National Encryption Policy in November 2025 that requires government entities to plan their transition to PQC.

This paper examines what the transition means for Saudi and GCC critical infrastructure. It combines a review of international standards, regional regulation and the regional threat history with two original, reproducible analyses. First, the NIST post-quantum standards (ML-KEM, ML-DSA and SLH-DSA), Falcon and Classic McEliece were benchmarked against classical baselines on an Apple M2 Pro using liboqs 0.16.0 and OpenSSL 3.6.4. Second, the framing and serial transmission cost of PQC messages was computed for DNP3, Modbus, CAN FD and IEEE 802.15.4, protocols widely used in pipelines, substations and water networks.

On the processor tested, ML-KEM was faster than P-256 ECDH, and ML-DSA and Falcon verified signatures as fast as or faster than ECDSA; only SLH-DSA signing was dramatically slower (roughly 1,200 to 25,000 times ECDSA). Message size, however, imposes large costs on legacy links: one ML-DSA-44 signature requires 10 DNP3 frames and about 3.0 seconds at 9,600 bit/s, against 0.09 seconds for ECDSA. For capable hardware, bandwidth, protocol framing and embedded firmware trust anchors are therefore more pressing constraints than computation. Building on these findings, the paper analyses implications for oil and gas, power, desalination, banking and government services in the region, maps PQC migration activities onto NCA and SAMA frameworks, and proposes a phased migration roadmap and algorithm selection guide for Saudi and GCC operators, regulators, vendors and cybersecurity assessors.

Keywords: post-quantum cryptography; critical infrastructure; Saudi Arabia; GCC; NCA; OTCC; National Cryptographic Standards; operational technology; ML-KEM; ML-DSA; DNP3; migration planning

Contents

Abstract	i
1 Introduction	1
1.1 The quantum threat	1
1.2 The standards and regulatory response	1
1.3 Why critical infrastructure is difficult	2
1.4 Research questions	2
1.5 Research gap	2
1.6 Contributions	2
1.7 Scope and organization	3
2 Background	4
2.1 The quantum threat to public-key cryptography	4
2.2 Post-quantum algorithm families	4
2.3 Standards status	5
2.4 Hybrid deployment and cryptographic agility	5
3 Regional Context: Saudi Arabia and the GCC	7
3.1 Critical infrastructure landscape	7
3.2 Threat history	7
3.3 Regulatory landscape	8
3.4 Implications for this study	9
4 Critical Infrastructure Environments and Constraints	10
4.1 Timing	10
4.2 Message size and bandwidth	10
4.3 Existing security layers	10
4.4 Lifetimes, trust anchors and supply chains	11
4.5 Validation and regulation	11
5 Methodology	12
5.1 Review	12
5.2 Benchmarks	12
5.3 Protocol framing analysis	13
6 Results	14
6.1 Key establishment	14
6.2 Digital signatures	15
6.3 Message sizes	16

6.4	Protocol framing and transmission time	17
6.5	Summary of findings	18
7	Sector Implications for Saudi Arabia and the GCC	19
7.1	Oil, gas and petrochemicals	19
7.2	Electric power and the GCC interconnection	19
7.3	Desalination and water transmission	19
7.4	Banking and payments	20
7.5	Government and national digital services	20
7.6	Cross-sector summary	20
8	Mapping PQC Migration to Regional Frameworks	20
8.1	Activity-to-framework mapping	21
8.2	Guidance for cybersecurity assessors	21
9	Migration Roadmap and Algorithm Selection	22
9.1	International timelines	22
9.2	Phased migration roadmap	23
9.3	Prioritizing assets with Mosca’s inequality	24
9.4	Algorithm selection guide	24
10	Discussion	26
10.1	Computation is not the main barrier on capable hardware	26
10.2	Size, framing and bandwidth are the dominant costs	26
10.3	The urgency is different for confidentiality and for authenticity	26
10.4	Regional considerations	27
10.5	Implementation risk	27
10.6	Economic considerations	27
10.7	Comparison with common assumptions	27
11	Recommendations	29
11.1	For critical infrastructure operators in Saudi Arabia and the GCC	29
11.2	For regulators (NCA, SAMA and GCC counterparts)	29
11.3	For vendors and integrators serving the region	29
11.4	For cybersecurity assessors and GRC practitioners	30
11.5	For standards bodies and researchers	30
12	Limitations	31
13	Conclusion and Future Work	32
13.1	Future work	32

References	33
A Reproducing the Benchmarks	36
B DNP3 Framing: Worked Example	37
C Glossary	38

List of Figures

1	Research design.	12
2	Key-establishment operations on Apple M2 Pro (log scale). ECDH has a single derivation step, shown in the encapsulation position.	14
3	Signing and verification time on Apple M2 Pro (log scale).	16
4	Lower-bound time to transmit one signature over a DNP3 serial link (log scale). . .	18
5	Published migration milestones on a common timeline (simplified from Table 12). Dots mark the Saudi NCS-2 consultation (Dec. 2025) and the UAE National Encryption Policy (Nov. 2025); NCS-1 has addressed PQC in an appendix since 2020. .	23
6	Proposed phased migration roadmap for Saudi and GCC CI operators, aligned with international milestones and regional frameworks.	23
7	Algorithm selection guide derived from the results. For per-message protection on very constrained or time-critical links, prefer symmetric MACs keyed through a post-quantum key exchange.	25

List of Tables

1	Algorithms considered in this study. NIST security categories: 1 $\approx$ AES-128, 3 $\approx$ AES-192, 5 $\approx$ AES-256.	5
2	Publicly documented cyber attacks on Saudi critical organizations relevant to cryptographic integrity.	8
3	Regional regulatory instruments relevant to PQC migration (status as understood at the time of writing; the Arabic versions of Saudi instruments are binding).	8
4	Characteristics of enterprise IT and OT relevant to cryptographic migration (summarized from NIST SP 800-82r3 [21]).	10
5	Key-establishment performance, mean time per operation in μs (population standard deviation in parentheses). Classical values derived from OpenSSL throughput (no dispersion reported).	14
6	Signature performance, mean time per operation in μs (population standard deviation in parentheses). SLH-DSA-128f and 128s values are from the confirmation run; the other SLH-DSA parameter sets are from the initial run (see Section 5).	15
7	Cost relative to ECDSA P-256 (values below 1 are faster than ECDSA).	16
8	Sizes in bytes, from the standards and specifications [8–10, 22, 23]. Falcon signatures vary in length; typical and maximum values are shown.	17
9	Minimum frames per message (lower bounds, headers ignored) and DNP3 serial transmission time (link-layer framing included; application headers and turnaround excluded).	17
10	Qualitative summary of where post-quantum costs are expected to bind in Saudi and GCC sectors (derived from the analysis in this section; not measured).	20
11	Mapping of PQC migration activities to areas of NCA and SAMA frameworks (proposed; not an official interpretation).	21
12	Published post-quantum migration milestones, international and regional.	22
13	Illustrative application of Mosca’s inequality to CI assets.	24
14	Common assumptions compared with the evidence from this study.	28

1 Introduction

Saudi Arabia and the other member states of the Gulf Cooperation Council (GCC), namely Bahrain, Kuwait, Oman, Qatar and the United Arab Emirates, operate some of the largest and most strategically important industrial infrastructure in the world. Oil and gas production and processing, petrochemicals, electric power generation and transmission, and seawater desalination all depend on industrial control systems (ICS) and other operational technology (OT). Saudi Arabia alone is the world’s largest producer of desalinated water, with capacity exceeding 16 million cubic metres per day by the end of 2025 [44]. Under Saudi Vision 2030, these sectors are being digitized rapidly [45], which increases both their efficiency and their dependence on cryptography.

The region’s infrastructure has also been a prominent target. The Shamoon malware destroyed data on around 30,000 workstations at Saudi Aramco in 2012 [46], and new Shamoon variants struck Saudi organizations again in 2016 and 2017 [47]. In 2017, the Triton (also called Trisis) malware targeted Triconex safety instrumented systems at a Saudi petrochemical facility, the first publicly known attack designed to manipulate industrial safety controllers [48, 49]; in 2022, the U.S. Department of Justice charged an employee of a Russian government research institute in connection with that intrusion [50]. These incidents show that capable state-level adversaries have both the interest and the patience to target Gulf industrial systems, which is precisely the threat model behind “harvest now, decrypt later” collection of encrypted traffic for future quantum decryption.

1.1 The quantum threat

The public-key cryptography that protects these systems, principally RSA and elliptic-curve cryptography (ECC¹), rests on mathematical problems that Shor’s algorithm solves in polynomial time on a sufficiently large fault-tolerant quantum computer [1]. No such cryptographically relevant quantum computer (CRQC) exists today, and estimates of when one might appear vary widely [3]. Resource estimates have fallen sharply, however: Gidney and Ekerå estimated in 2021 that RSA-2048 could be factored in about eight hours with 20 million noisy qubits [4], and Gidney later revised the requirement to fewer than one million noisy qubits running for under a week [5]. Quantum computing is also arriving in the region itself: Aramco and Pasqal deployed Saudi Arabia’s first quantum computer, a 200-qubit neutral-atom system for industrial research, in Dhahran in November 2025 [51]. Such machines are far from able to break cryptography, but they reflect the pace of investment.

1.2 The standards and regulatory response

NIST published its first PQC standards, ML-KEM (FIPS 203), ML-DSA (FIPS 204) and SLH-DSA (FIPS 205), in August 2024 [8–10], and its draft transition guidance proposes deprecating quantum-vulnerable public-key algorithms after 2030 and disallowing them after 2035 [12]. The UK and the EU have set similar milestones, with the EU asking that high-risk use cases be migrated by the end of 2030 [15, 17].

The Gulf has begun to respond. Saudi Arabia’s National Cybersecurity Authority (NCA) included an appendix on post-quantum cryptography in its National Cryptographic Standards (NCS-1:2020) [52], and in December 2025 consulted publicly on NCS-2:2025, a draft update that

¹In this paper, “ECC” alone denotes elliptic-curve cryptography; the NCA’s Essential Cybersecurity Controls are always written “NCA ECC” or “ECC-2:2024”.

adds standards for post-quantum algorithms and requirements on quantum key distribution [53]. In November 2025, the UAE approved a National Encryption Policy and executive regulation that call on government entities to prepare approved transition plans from traditional encryption to post-quantum cryptography [58], supported by a National Post-Quantum Migration Programme [59].

1.3 Why critical infrastructure is difficult

OT prioritizes availability and deterministic timing, runs on long-lived and resource-constrained devices, and uses industrial protocols designed for small messages and slow links [21]. A change of cryptographic algorithm that is routine in a web browser can require hardware replacement, protocol redesign, recertification and a planned outage at a refinery, substation or desalination plant. In the Gulf, the scale and national importance of these facilities, their reliance on international vendors, and the regulatory requirements of the NCA’s Operational Technology Cybersecurity Controls (OTCC) [55] add further constraints.

1.4 Research questions

1. What are the main technical constraints on deploying standardized PQC in CI environments of the kind operated in Saudi Arabia and the GCC?
2. How large are the computational and communication costs of PQC relative to the classical algorithms it replaces, and which of these costs is most likely to limit deployments?
3. What migration priorities, regulatory alignment and practical strategies follow from this evidence for Saudi and GCC operators and regulators?

1.5 Research gap

Much of the PQC literature evaluates algorithms in isolation or in Internet protocols such as TLS [38, 39], and CI security guidance treats cryptography in general terms [21]. Published analyses of PQC migration are also written almost entirely from U.S. and European perspectives, with regulatory recommendations framed around NIST, NCSC and EU instruments. Little work connects measured PQC costs to the framing, bandwidth and lifecycle constraints of industrial protocols, and the author is not aware of published work that does so for the sectors, threat history and regulatory frameworks of Saudi Arabia and the GCC.

1.6 Contributions

- (i) A review of PQC standards and international timelines alongside the Saudi and UAE regulatory landscape and the regional threat history.
- (ii) Reproducible benchmarks of the FIPS PQC standards, Falcon and Classic McEliece against ECDSA, ECDH and RSA, with raw data, scripts and commands provided (Appendix A).
- (iii) A protocol framing analysis quantifying how PQC message sizes translate into frames and transmission time on common industrial links.
- (iv) A sector analysis for oil and gas, power, desalination, banking and government services in the region.

- (v) A mapping of PQC migration activities onto NCA and SAMA frameworks, including guidance for cybersecurity assessors.
- (vi) A phased migration roadmap, algorithm selection guide and recommendations for regional operators, regulators, vendors and assessors.

1.7 Scope and organization

This is a single-author study based on public information and the author’s own measurements. It does not include surveys, interviews, field measurements or any confidential information from operators, and its benchmarks cover a single desktop-class processor rather than embedded controllers. These limits are discussed in Section 12.

Section 2 reviews the quantum threat and post-quantum algorithms. Section 3 describes the regional context. Section 4 describes the technical constraints of CI environments, Section 5 the methodology, and Section 6 the results. Section 7 analyses sector implications, Section 8 maps migration activities to regional frameworks, and Section 9 proposes a roadmap. Sections 10 to 13 discuss the findings and give recommendations, limitations and conclusions.

2 Background

2.1 The quantum threat to public-key cryptography

Shor’s 1994 algorithm factors integers and computes discrete logarithms in polynomial time on a quantum computer [1]. It therefore breaks RSA, finite-field Diffie–Hellman, DSA, ECDSA, ECDH and EdDSA, which together account for nearly all public-key cryptography deployed in CI today. The attack requires a large, error-corrected machine; present devices are far from that scale, and running Shor’s algorithm against RSA-2048 is a matter of engineering progress rather than a known imminent capability.

Grover’s algorithm offers a quadratic speed-up for unstructured search [2]. In principle this halves the effective security of symmetric keys, but the attack parallelizes poorly and requires long sequential quantum computation, so AES-128 is widely regarded as remaining practically secure, and AES-256 provides a large margin. NIST’s transition guidance keeps AES-128, AES-192 and AES-256 approved [12]. Hash functions such as SHA-256 and SHA-3 are similarly affected only modestly.

Resource estimates. The number of physical qubits needed to break RSA-2048 has fallen with each generation of algorithmic and error-correction improvements. Gidney and Ekerå estimated 20 million noisy qubits and about eight hours [4]; Gidney’s 2025 estimate is below one million noisy qubits and under a week of runtime [5]. These are estimates for hypothetical hardware, not demonstrations, but they explain why agencies have hardened their timelines.

Harvest now, decrypt later. An adversary can record encrypted traffic today and decrypt it once a CRQC exists. Any data whose confidentiality must outlast the arrival of a CRQC is already at risk if it is protected by quantum-vulnerable key establishment. Signatures are different: a recorded signature cannot be forged retroactively, so authentication is threatened only from the moment a CRQC becomes available. This distinction shapes migration priorities (Section 10).

Mosca’s inequality. Mosca expresses migration urgency as the inequality $x + y > z$, where x is how long data must remain secure, y is the time needed to migrate, and z is the time until a CRQC exists [6]. If the inequality holds, some data will be exposed. In CI, y is large because equipment is replaced slowly, and for authenticity the relevant x is effectively the remaining service life of deployed devices.

2.2 Post-quantum algorithm families

Post-quantum cryptography comprises classical algorithms, running on ordinary computers, whose security is believed to hold against quantum attackers [7]. The main families are:

Lattice-based. Security rests on problems such as Learning With Errors (LWE) and its structured variants. ML-KEM (formerly CRYSTALS-Kyber) and ML-DSA (formerly CRYSTALS-Dilithium) use module lattices [8, 9]; Falcon (FN-DSA) uses NTRU lattices and fast Fourier sampling to produce compact signatures [22]. Lattice schemes offer the best overall balance of speed and size and form the core of the new standards.

Hash-based. Signatures built only from hash functions, whose security is very well understood. SLH-DSA (formerly SPHINCS+) is stateless [10]; LMS and XMSS are stateful and require the signer never to reuse a one-time key, which suits controlled environments such as firmware signing [24]. Hash-based signatures are large and, in the stateless case, slow to sign.

Code-based. Security rests on decoding random linear codes. Classic McEliece has resisted cryptanalysis since 1978 but has public keys of hundreds of kilobytes to over a megabyte [23]. HQC, selected by NIST in 2025, is a code-based KEM with much smaller keys, intended as a backup to ML-KEM based on a different mathematical assumption [11].

Multivariate and isogeny-based. These families produced some of the smallest keys or signatures among candidates, but prominent schemes were broken during standardization: the multivariate signature Rainbow by Beullens in 2022 [25], and the isogeny-based KEM SIKE by Castryck and Decru [26]. Both breaks occurred late in a multi-year public evaluation, which is a strong argument for cryptographic agility and for hybrid deployment during the transition.

2.3 Standards status

Table 1 summarizes the algorithms considered in this study and their status.

Table 1: Algorithms considered in this study. NIST security categories: 1 $\approx$ AES-128, 3 $\approx$ AES-192, 5 $\approx$ AES-256.

Algorithm	Role	Basis	Status (Sept. 2026)
ML-KEM-512/768/1024	KEM	Module lattices	FIPS 203 (final, 2024)
ML-DSA-44/65/87	Signature	Module lattices	FIPS 204 (final, 2024)
SLH-DSA (128–256, s/f)	Signature	Hash functions	FIPS 205 (final, 2024)
Falcon-512/1024 (FN-DSA)	Signature	NTRU lattices	FIPS 206 (draft)
Classic McEliece-348864	KEM	Binary Goppa codes	Not selected by NIST
HQC	KEM	Quasi-cyclic codes	Selected 2025; draft pending
LMS / XMSS	Signature (stateful)	Hash functions	SP 800-208 (final, 2020)
ECDSA / ECDH P-256	Baseline	Elliptic curves	Deprecated after 2030 (proposed)
RSA-2048	Baseline	Factoring	Deprecated after 2030 (proposed)

2.4 Hybrid deployment and cryptographic agility

During the transition, hybrid schemes combine a classical and a post-quantum algorithm so that security holds as long as either remains unbroken. Hybrid key exchange is already widely deployed on the Internet through the X25519MLKEM768 TLS 1.3 key share [27, 28], and IR 8547 notes that hybrid use is not affected by the proposed 2035 disallowance of classical-only algorithms [12]. Several European agencies, including Germany’s BSI and France’s ANSSI, recommend hybrid deployment during the transition [19, 20], while the NSA’s CNSA 2.0 suite for U.S. national security systems moves directly to specified post-quantum algorithms [13].

Cryptographic agility, the ability to replace algorithms without redesigning systems, was identified early as a precondition for migration [29] and was a design goal of the Open Quantum Safe project [30]. Agility requires algorithm identifiers and negotiation in protocols, room for larger

keys and signatures in message formats and storage, and update mechanisms that can deliver new cryptographic code to devices in the field.

3 Regional Context: Saudi Arabia and the GCC

3.1 Critical infrastructure landscape

Oil, gas and petrochemicals. Hydrocarbon production, processing and export, together with petrochemical manufacturing, form the core of GCC economies. These facilities rely on distributed control systems (DCS) inside plants, safety instrumented systems (SIS) that shut processes down when conditions become dangerous, and supervisory control and data acquisition (SCADA) systems spanning wellheads, pipelines and export terminals, often connected over radio, microwave or satellite links. Plants are designed for decades of operation.

Electric power. Utilities operate generation, transmission and distribution networks with substation automation based on IEC 61850 and telecontrol protocols such as IEC 60870-5-104 and DNP3. The six GCC states are linked through a regional grid interconnection operated by the GCC Interconnection Authority [60], so some control and protection functions depend on coordination across national boundaries.

Desalination and water. Seawater desalination supplies a large share of drinking water in the region. Saudi Arabia is the world’s largest producer, with capacity exceeding 16 million cubic metres per day at the end of 2025 [44], delivered through long transmission pipelines with remote pumping and storage stations. Desalination plants are large industrial facilities; the pipelines and remote stations resemble the widely dispersed, low-bandwidth SCADA networks of the water sector elsewhere.

Banking and payments. Gulf banking systems are highly digitized and regulated by central banks; in Saudi Arabia, the Saudi Central Bank (SAMA) Cyber Security Framework sets requirements for member organizations [57].

Government and digital services. Vision 2030 has driven extensive digitization of government services [45], which rely on public-key infrastructure for identity, signatures and secure communication, and which hold records with long confidentiality requirements.

3.2 Threat history

Table 2 summarizes publicly documented attacks on Saudi industrial and critical organizations that are relevant to this study.

Table 2: Publicly documented cyber attacks on Saudi critical organizations relevant to cryptographic integrity.

Incident	Year	Relevance to this study
Shamoon [46]	2012	Destructive wiper attack on around 30,000 Saudi Aramco workstations; showed that a state-level adversary would target the region’s largest energy company.
Shamoon 2 [47]	2016–17	Renewed destructive attacks on Saudi government and private organizations; confirmed a persistent adversary interest.
Triton / Trisis [48–50]	2017	Malware designed to reprogram Triconex safety controllers at a Saudi petrochemical facility. It demonstrated that attackers can target the logic and firmware of safety-critical controllers, whose integrity depends on reliable authentication of code.

Two lessons follow for PQC. First, adversaries with the resources and patience of those behind these attacks are exactly those most likely to collect encrypted traffic today for future quantum decryption. Second, Triton showed that the integrity of controller logic and firmware is itself a safety issue. The cryptographic mechanisms that protect firmware and logic updates will increasingly be digital signatures, and if those signatures are ECDSA or RSA only, a future CRQC would allow an adversary to forge authentic-looking updates.

3.3 Regulatory landscape

Table 3 summarizes the Saudi and UAE instruments most relevant to PQC migration in critical infrastructure.

Table 3: Regional regulatory instruments relevant to PQC migration (status as understood at the time of writing; the Arabic versions of Saudi instruments are binding).

Instrument	Relevance
NCA Essential Cybersecurity Controls, ECC-2:2024 [54]	Mandatory baseline for government entities and operators of critical national infrastructure (CNI); includes cryptography requirements. OT requirements are addressed in the separate OTCC.
NCA Operational Technology Cybersecurity Controls, OTCC-1:2022 [55]	Extension of NCA ECC for industrial control systems in CNI: four domains (governance, defence, resilience, third-party), 47 main controls and 122 subcontrols, applied at three levels according to facility criticality.
NCA Critical Systems Cybersecurity Controls, CSCC-1:2019 [56]	Additional requirements for systems designated as critical, including cryptography.
NCA National Cryptographic Standards, NCS-1:2020 [52]	Minimum cryptographic requirements at MODERATE and ADVANCED levels; appendices address post-quantum cryptography and side-channel attacks. Referenced by NCA ECC, CCC and CSCC cryptography controls.
NCA NCS-2:2025 (consultation draft, Dec. 2025) [53]	Draft successor to NCS-1 that adds standards for post-quantum algorithms, requirements on quantum key distribution, and other updates.
SAMA Cyber Security Framework [57]	Cybersecurity requirements for banks, insurers and other SAMA-regulated organizations, including cryptography.
UAE National Encryption Policy (Nov. 2025) [58]	Requires UAE government entities to develop approved transition plans from traditional encryption to PQC, overseen by the UAE Cybersecurity Council.
UAE National Post-Quantum Migration Programme [59]	National programme to identify vulnerable systems and support entities through PQC deployment.

Three features of this landscape shape the analysis that follows. First, Saudi regulation already routes cryptographic choices through a single national standard, NCS, which is referenced by the NCA ECC, CCC and CSCC cryptography controls [52]; once NCS-2 is finalized, adding post-quantum algorithms to the approved set will propagate through these frameworks. Second, OT in critical infrastructure is governed by OTCC, which applies graded requirements according to facility criticality; this provides a natural vehicle for OT-specific PQC requirements. Third, the UAE has made transition planning mandatory for government entities, while in Saudi Arabia no dated migration deadline had been published at the time of writing. The author did not identify comparable published PQC migration mandates for the other GCC states.

3.4 Implications for this study

The regional context raises the stakes of three findings developed later in this paper. The dominance of long-lived OT in Gulf economies means that equipment bought today will operate well past any plausible migration deadline. The region’s reliance on SCADA over dispersed, often low-bandwidth links makes message size a central constraint. And the Triton precedent makes the post-quantum protection of firmware and controller logic a safety concern, not only a security one.

4 Critical Infrastructure Environments and Constraints

NIST SP 800-82r3 summarizes the characteristics that distinguish OT from enterprise IT [21]. Table 4 restates those most relevant to cryptographic migration.

Table 4: Characteristics of enterprise IT and OT relevant to cryptographic migration (summarized from NIST SP 800-82r3 [21]).

Aspect	Enterprise IT	Operational technology
Priority	Confidentiality and integrity	Availability, safety and integrity
Timing	Delays usually tolerable	Deterministic, sometimes millisecond deadlines
Lifetime	3–5 years	Often 15–20 years or more
Patching	Frequent, automated	Rare, scheduled around outages; may require recertification
Hardware	General-purpose, ample resources	Embedded controllers with limited CPU, memory and entropy
Networks	High-bandwidth IP	Mix of IP and low-bandwidth serial, radio and fieldbus links
Change control	Vendor-driven updates	Vendor, integrator and regulator involvement

Five constraints are particularly relevant to PQC.

4.1 Timing

Protection signalling in substations is time-critical: IEC 61850-5 defines transfer-time classes as short as 3 ms for trip messages [31]. Per-message public-key signatures are difficult to fit in such budgets even with classical algorithms, which is one reason IEC 62351-6 protection of GOOSE and sampled values relies primarily on message authentication codes rather than signatures [32]. Control loops in process industries and interlocking logic in rail signalling have similar deterministic requirements. Any cryptographic operation in the critical path must have a bounded worst-case execution time, not merely a good average, which matters for algorithms such as ML-DSA whose signing time varies because of rejection sampling.

4.2 Message size and bandwidth

Legacy industrial protocols were designed for small payloads. A Modbus protocol data unit is limited to 253 bytes [33]; a DNP3 link-layer frame carries at most 250 user-data octets within a 292-octet frame [34]; a CAN FD frame carries 64 bytes; and an IEEE 802.15.4 frame, used by WirelessHART and ISA100.11a field networks, is limited to 127 bytes including headers. Many remote sites still communicate over serial or licensed radio links at a few thousand to tens of thousands of bits per second. Section 6 quantifies how PQC sizes interact with these limits.

4.3 Existing security layers

Where industrial protocols are secured, they use a mix of symmetric and public-key mechanisms. DNP3 Secure Authentication (IEEE 1815-2012) authenticates messages with HMACs and uses

public-key cryptography only in some methods for changing update keys [34]. IEC 62351 secures IEC 60870-5-104 and IEC 61850 client-server traffic with TLS profiles (IEC 62351-3) and uses MACs for multicast GOOSE and sampled values [32]. The Modbus/TCP Security specification wraps Modbus in TLS with X.509 certificates [35]. These choices matter because symmetric MACs do not need to change for quantum resistance, while TLS handshakes, certificate chains and public-key key-update methods do.

4.4 Lifetimes, trust anchors and supply chains

Controllers, relays and remote terminal units often remain in service for 15 to 20 years or more [21]. Many devices embed a manufacturer’s public key in read-only or one-time-programmable memory to verify firmware at boot. If that key is ECDSA or RSA only, the device cannot be migrated to post-quantum firmware authentication later without hardware replacement. CI equipment also reaches operators through long supply chains of component vendors, integrators and service providers, so post-quantum capability depends on cryptographic libraries, secure elements and hardware security modules chosen years earlier by others.

4.5 Validation and regulation

CI operators typically require cryptographic modules validated under FIPS 140-3 or equivalent schemes, and sector regulations such as NERC CIP in North American electricity [36] or the security programme requirements of ISA/IEC 62443 [37] shape which products can be deployed. None of these currently mandates specific post-quantum algorithms. Validation programmes for the new FIPS algorithms are emerging, and certified products lag behind the standards, which lengthens the y term in Mosca’s inequality for regulated operators.

5 Methodology

The study combines a structured review with two quantitative analyses. Figure 1 summarizes the research design.

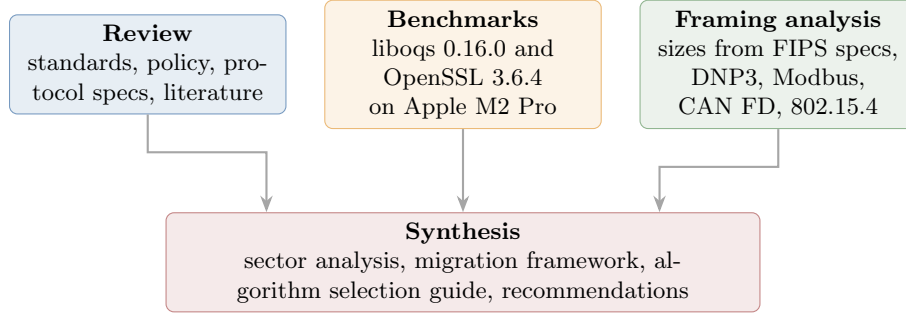


Figure 1: Research design.

5.1 Review

Primary sources were preferred: NIST FIPS and special publications; government guidance from the NCSC, EU and NSA; Saudi NCA and SAMA frameworks and the NCS-2 consultation draft; official UAE announcements; incident analyses by security vendors and the U.S. Department of Justice; IETF drafts; IEC and IEEE protocol standards; and peer-reviewed or widely cited cryptographic literature. Where Saudi instruments are published in Arabic and English, the English versions were used; the Arabic versions are binding. Algorithm names and statuses were checked against the published standards. Secondary sources were used only to confirm the status of documents not yet final (FN-DSA and IR 8547).

5.2 Benchmarks

Platform. Apple M2 Pro, macOS 26.5.1 (build 25F80). Post-quantum algorithms were measured with the `speed_kem` and `speed_sig` tools of liboqs 0.16.0 (commit b196b57), compiled with Apple clang 21 in its default distribution configuration (`OQS_DIST_BUILD`, `CMAKE_BUILD_TYPE=Release`, OpenSSL 3.6.4 providing AES and SHA-2) [43]. Classical baselines were measured with `openssl speed` from OpenSSL 3.6.4 (Homebrew build).

Algorithms. ML-KEM-512, 768 and 1024; Classic McEliece-348864; ML-DSA-44, 65 and 87; Falcon-512 and 1024; and the six SHA-2 parameter sets of SLH-DSA. Classical baselines were ECDSA and ECDH over P-256 and RSA-2048.

Procedure. liboqs repeats each operation for approximately 3 seconds and reports the number of iterations, mean time and population standard deviation. OpenSSL repeats each operation for 10 seconds and reports throughput, from which mean time per operation was derived; it does not report dispersion. The machine was on mains power with other applications closed. Each algorithm was run once. The “CPU cycles” column reported by liboqs was not used, because on this platform it reflects nanoseconds rather than processor cycles.

Data quality note. An initial run of the complete signature suite produced an output file that was partially overwritten; its SLH-DSA section was intact. SLH-DSA-SHA2-128f and 128s were then re-run individually as a confirmation. The two runs agreed to within 1.3% for every operation (for example, 128f signing: 27.55 ms and 27.75 ms; 128s verification: 555.6 μ s and 556.6 μ s), which supports the reliability of the initial run. The confirmation-run values are reported for 128f and 128s, and initial-run values for the remaining SLH-DSA parameter sets. All other post-quantum results come from single-algorithm runs.

5.3 Protocol framing analysis

For each message (a signature, an ML-KEM ciphertext or an encapsulation key), the minimum number of frames needed to carry it was computed for CAN FD (64-byte payload), IEEE 802.15.4 (127-byte frame), a Modbus PDU (253 bytes) and DNP3. These counts ignore protocol headers and are therefore lower bounds.

For DNP3 over a serial link, the exact on-wire size was computed from the link-layer format. Each frame carries up to 249 application octets plus a 1-octet transport header, and a frame with n user-data octets occupies

$$L(n) = 10 + n + 2 \left\lceil \frac{n}{16} \right\rceil \quad \text{octets}, \quad (1)$$

consisting of a 10-octet header block and a 2-octet CRC after every 16 data octets. A full frame ($n = 250$) therefore occupies 292 octets. Transmission time assumes 10 bits per octet (8 data bits, start and stop bits) at 9,600 and 19,200 bit/s, and excludes application-layer headers, turnaround and retransmissions, so it too is a lower bound. A worked example is given in Appendix B. Message sizes were taken from the standards (Table 8).

6 Results

6.1 Key establishment

Table 5 and Figure 2 report the key-establishment results. All three ML-KEM parameter sets completed every operation in under $33\ \mu\text{s}$. Encapsulation with ML-KEM-768, the parameter set used in the widely deployed TLS hybrid, took $18.3\ \mu\text{s}$, about one third of the $52.6\ \mu\text{s}$ needed for a single P-256 ECDH shared-secret derivation. A complete ML-KEM-768 exchange (key generation and decapsulation on one side, encapsulation on the other) cost about $57\ \mu\text{s}$ of total computation, less than the two ECDH derivations (about $105\ \mu\text{s}$) required in an elliptic-curve exchange, even before counting ECDH key generation.

Classic McEliece-348864 behaved very differently: encapsulation was fast ($23.1\ \mu\text{s}$), but decapsulation took $24.2\ \text{ms}$ and key generation averaged $256\ \text{ms}$ with a standard deviation of $188\ \text{ms}$ over 12 runs. The high variance reflects the algorithm’s rejection-based key generation. Classic McEliece is therefore suited to long-lived static keys rather than ephemeral exchanges, and its $261\ \text{KB}$ public key (Table 8) rules it out for most constrained links.

Table 5: Key-establishment performance, mean time per operation in μs (population standard deviation in parentheses). Classical values derived from OpenSSL throughput (no dispersion reported).

Algorithm	Key generation	Encapsulation	Decapsulation
ECDH P-256 (shared-secret derivation)	–	52.56 per derivation	
RSA-2048 (as KEM)	53,191	20.69	725.27
ML-KEM-512	10.44 (8.12)	11.62 (1.78)	13.80 (3.82)
ML-KEM-768	17.38 (7.21)	18.25 (2.30)	21.37 (4.18)
ML-KEM-1024	26.13 (5.57)	28.08 (5.02)	32.30 (8.46)
Classic McEliece-348864	255,753 (188,185)	23.13 (7.14)	24,248 (222)

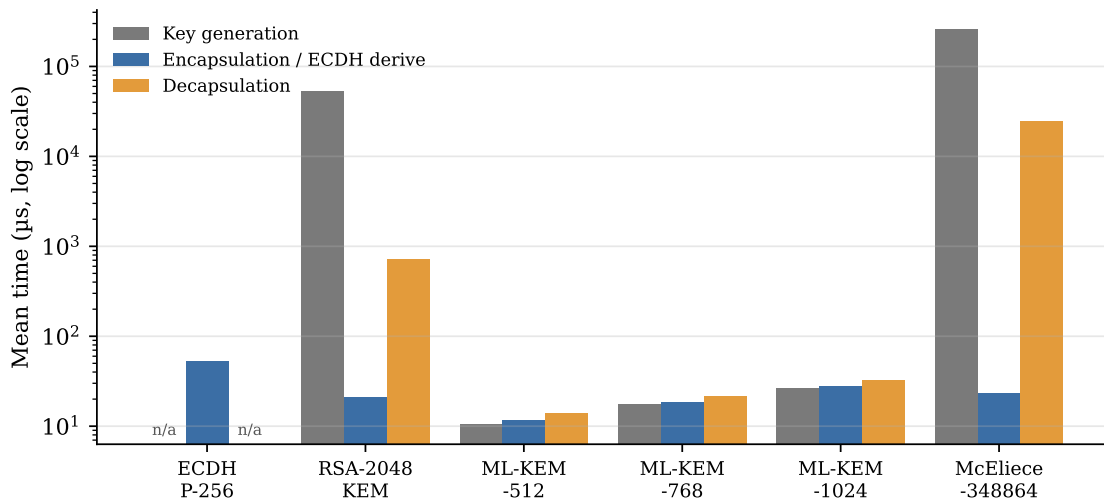


Figure 2: Key-establishment operations on Apple M2 Pro (log scale). ECDH has a single derivation step, shown in the encapsulation position.

6.2 Digital signatures

Table 6 and Figure 3 report the signature results, and Table 7 expresses them relative to ECDSA P-256.

Lattice signatures were competitive with classical algorithms. ML-DSA-44 signed in $94.5 \mu\text{s}$ (4.1 times ECDSA) and verified in $37.6 \mu\text{s}$, faster than ECDSA verification ($69.6 \mu\text{s}$). ML-DSA signing times varied considerably (standard deviation of more than half the mean) because signing uses rejection sampling and repeats a random number of times; verification times were stable. Falcon-512 verified fastest of all algorithms tested ($31.1 \mu\text{s}$) and signed in $204 \mu\text{s}$, but its key generation took about 6 ms. Every lattice parameter set signed faster than RSA-2048 ($727 \mu\text{s}$).

SLH-DSA was the clear outlier. The fast variant, SLH-DSA-SHA2-128f, signed in 27.8 ms (about 1,200 times ECDSA) and verified in 1.68 ms (about 24 times). The small variant, 128s, verified faster (0.56 ms, about 8 times ECDSA) but took 579 ms to sign. For SLH-DSA, the bottleneck is signing, not verification. The “s” variants trade signing time for smaller signatures and faster verification, which is the right trade-off when signing happens rarely and offline.

Table 6: Signature performance, mean time per operation in μs (population standard deviation in parentheses). SLH-DSA-128f and 128s values are from the confirmation run; the other SLH-DSA parameter sets are from the initial run (see Section 5).

Algorithm	Key generation	Sign	Verify
ECDSA P-256	–	22.98	69.55
RSA-2048	56,818	727.38	18.91
ML-DSA-44	40.09 (2.39)	94.49 (51.31)	37.56 (1.80)
ML-DSA-65	69.50 (5.79)	145.56 (75.66)	63.97 (3.22)
ML-DSA-87	119.67 (8.05)	205.37 (83.14)	112.08 (3.76)
Falcon-512	6,016 (1,420)	204.03 (6.17)	31.13 (1.56)
Falcon-1024	19,590 (5,764)	409.24 (18.14)	61.35 (2.66)
SLH-DSA-SHA2-128f	1,195 (50)	27,753 (950)	1,676 (27)
SLH-DSA-SHA2-128s	75,880 (289)	578,612 (1,504)	557 (21)
SLH-DSA-SHA2-192f	1,720 (136)	45,038 (304)	2,399 (38)
SLH-DSA-SHA2-192s	109,262 (750)	1,010,849 (1,143)	881 (16)
SLH-DSA-SHA2-256f	4,536 (49)	92,787 (507)	2,529 (37)
SLH-DSA-SHA2-256s	72,413 (470)	898,592 (4,593)	1,215 (19)

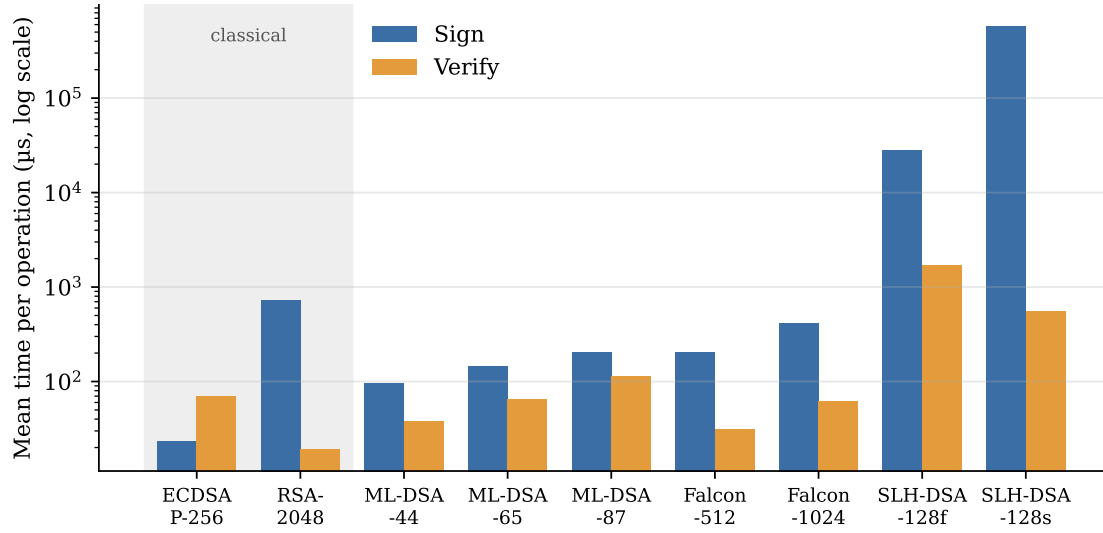


Figure 3: Signing and verification time on Apple M2 Pro (log scale).

Table 7: Cost relative to ECDSA P-256 (values below 1 are faster than ECDSA).

Algorithm	Sign / ECDSA sign	Verify / ECDSA verify
ML-DSA-44	4.1	0.54
ML-DSA-65	6.3	0.92
ML-DSA-87	8.9	1.61
Falcon-512	8.9	0.45
Falcon-1024	17.8	0.88
SLH-DSA-SHA2-128f	1,208	24.1
SLH-DSA-SHA2-128s	25,182	8.0

6.3 Message sizes

Table 8 lists key, ciphertext and signature sizes. Every standardized PQC signature is at least ten times larger than a 64-byte ECDSA signature, and ML-KEM messages are roughly 1 KB, compared with 32 to 65 bytes for elliptic-curve public keys. Certificate chains grow correspondingly: a chain of a leaf and one intermediate certificate carries two public keys and two signatures, so an ML-DSA-44 chain adds roughly 7 KB of key and signature material before any other certificate fields.

Table 8: Sizes in bytes, from the standards and specifications [8–10, 22, 23]. Falcon signatures vary in length; typical and maximum values are shown.

Algorithm	Public / encapsulation key	Signature or ciphertext
ECDSA / ECDH P-256	64 (65 uncompressed encoding)	64 (signature)
X25519	32	–
RSA-2048	256 (modulus)	256
ML-KEM-512 / 768 / 1024	800 / 1,184 / 1,568	768 / 1,088 / 1,568 (ciphertext)
Classic McEliece-348864	261,120	96 (ciphertext)
ML-DSA-44 / 65 / 87	1,312 / 1,952 / 2,592	2,420 / 3,309 / 4,627
Falcon-512 / 1024	897 / 1,793	666 (max 752) / 1,280 (max 1,462)
SLH-DSA-SHA2-128s / 128f	32	7,856 / 17,088
SLH-DSA-SHA2-192s / 192f	48	16,224 / 35,664
SLH-DSA-SHA2-256s / 256f	64	29,792 / 49,856

6.4 Protocol framing and transmission time

Table 9 converts these sizes into frames. An ECDSA signature fits in a single frame on every link considered. An ML-DSA-44 signature needs at least 38 CAN FD frames, 20 IEEE 802.15.4 frames, and 10 Modbus or DNP3 frames. On a 9,600 bit/s DNP3 serial link, the 2,842 on-wire octets take about 3.0 seconds to transmit, compared with 0.09 seconds for ECDSA (Figure 4). Even the compact Falcon-512 signature takes about 0.9 seconds, and an SLH-DSA-128f signature takes about 21 seconds.

The comparison with computation is striking. On the benchmark platform, producing and checking an ML-DSA-44 signature took about 0.13 ms; transmitting it over a 9,600 bit/s serial link takes over 20,000 times longer. Signatures larger than about 2 KB also exceed the application fragment size commonly configured for DNP3 outstations, so they would require multi-fragment messages, adding further round trips.

Table 9: Minimum frames per message (lower bounds, headers ignored) and DNP3 serial transmission time (link-layer framing included; application headers and turnaround excluded).

Message	Bytes	CAN FD	802.15.4	Modbus	DNP3	DNP3 serial time	
		(64 B)	(127 B)	(253 B)	frames	9,600 bit/s	19,200 bit/s
ECDSA P-256 signature	64	1	1	1	1	0.09 s	0.04 s
Falcon-512 signature (max)	752	12	6	3	4	0.93 s	0.47 s
ML-DSA-44 signature	2,420	38	20	10	10	2.96 s	1.48 s
ML-DSA-65 signature	3,309	52	27	14	14	4.05 s	2.03 s
ML-DSA-87 signature	4,627	73	37	19	19	5.66 s	2.83 s
SLH-DSA-128s signature	7,856	123	62	32	32	9.60 s	4.80 s
SLH-DSA-128f signature	17,088	267	135	68	69	20.88 s	10.44 s
ML-KEM-768 ciphertext	1,088	17	9	5	5	1.34 s	0.67 s
ML-KEM-768 encaps. key	1,184	19	10	5	5	1.45 s	0.72 s

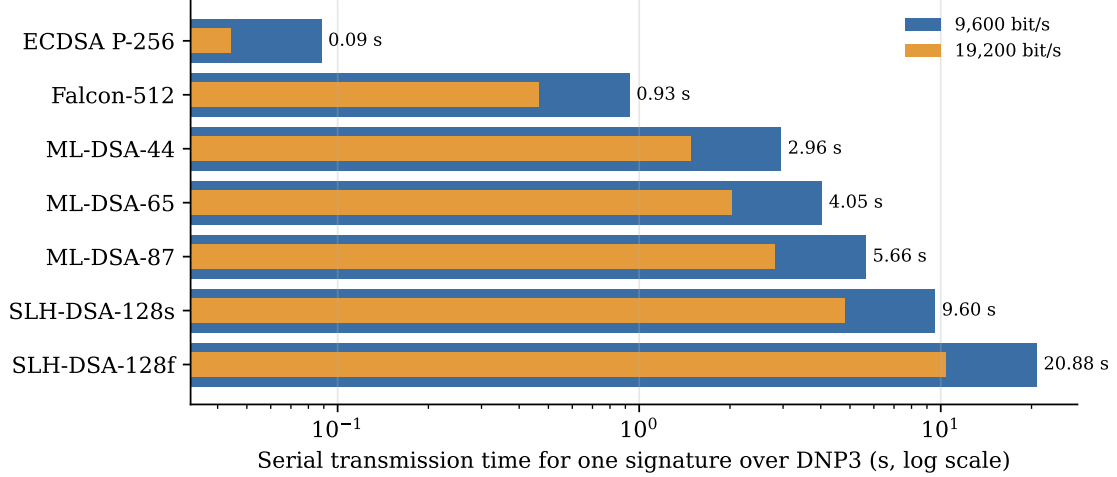


Figure 4: Lower-bound time to transmit one signature over a DNP3 serial link (log scale).

6.5 Summary of findings

1. On a capable processor, ML-KEM is faster than ECDH, and ML-DSA and Falcon verification is as fast as or faster than ECDSA.
2. Post-quantum signing is slower than ECDSA by a factor of about 4 to 18 for lattice schemes and by three to four orders of magnitude for SLH-DSA.
3. All post-quantum messages are much larger than their classical equivalents, and on low-bandwidth industrial links the resulting transmission time exceeds computation time by several orders of magnitude.
4. Classic McEliece and SLH-DSA are suited only to specific roles: static long-term keys and rarely signed artefacts, respectively.

7 Sector Implications for Saudi Arabia and the GCC

The measurements in Section 6 are sector-neutral. This section interprets them for the main CI sectors of Saudi Arabia and the GCC, drawing on published protocol standards, regulation and incident reports. It does not report data from operators; it identifies where, given each sector’s technology and constraints, the measured costs are likely to matter most. Table 10 summarizes the analysis.

7.1 Oil, gas and petrochemicals

Plant control systems and safety instrumented systems are the highest-consequence assets in the sector, and Triton showed that their logic and firmware are realistic targets [48]. Within plants, control networks are typically Ethernet based, so the framing costs in Table 9 matter less than device capability and change control. The priority is the integrity of controller logic and firmware: new DCS and SIS equipment should verify firmware and engineering changes with post-quantum or hybrid signatures, which is well matched to PQC because signing happens rarely and offline while verification is fast (Section 6). Safety certification of SIS makes such changes slow, so requirements must enter procurement long before deployment.

Outside plants, pipeline and wellhead SCADA often runs over radio, microwave or satellite links. Here the framing results apply directly: per-message post-quantum signatures would consume seconds of link time, so routine telemetry and control should remain protected by symmetric MACs, with public-key operations confined to key establishment and key updates, and gateways used at the boundary of legacy segments.

7.2 Electric power and the GCC interconnection

Substation protection messages (GOOSE) have millisecond deadlines and are protected with MACs under IEC 62351-6 [32], which do not need to change; what must migrate is the establishment and distribution of the keys behind them. Telecontrol links using TLS profiles can adopt hybrid ML-KEM with modest cost on IP networks, while serial links to distribution field devices face the framing costs quantified in Table 9. The regional interconnection adds a coordination requirement: where security associations cross national boundaries, interconnected utilities will need compatible algorithm choices and migration timing, which argues for regional coordination through the GCC Interconnection Authority and national regulators [60].

7.3 Desalination and water transmission

Desalination plants combine large process-control systems, similar to those of power and chemical plants, with long transmission pipelines and remote pumping and storage stations. The remote segments are where the framing analysis is most consequential: multi-second signature transfers would compete with operational data on narrow links, and older remote terminal units may have no firmware update path at all. Given the national importance of water supply in the region, the near-term strategy should be to protect remote links at gateways, secure remote-access paths with hybrid VPNs, apply OTCC segmentation controls [55], and require post-quantum-capable equipment in new projects, several of which are being built under Vision 2030 expansion plans [44].

7.4 Banking and payments

The banking sector relies heavily on TLS, hardware security modules and PKI, operates under the SAMA Cyber Security Framework [57], and handles data with multi-year confidentiality requirements. The benchmark results indicate that computation is not a barrier for server-side financial systems: ML-KEM and ML-DSA verification are fast, and ML-DSA signing is faster than the RSA-2048 signing widely used today. The main costs are hardware security module support, certificate and message sizes, and coordinated change with international payment networks. Banks are well placed to move early with hybrid key establishment on external and inter-bank links.

7.5 Government and national digital services

Government digital services depend on national PKI for identity, signatures and secure communication, and hold records whose confidentiality must last decades, making them highly exposed to harvest-now-decrypt-later collection. PKI roots have long lifetimes and long issuance lead times, so planning post-quantum or hybrid roots should begin early. The UAE’s requirement for government transition plans [58] provides one regional model for this sector.

7.6 Cross-sector summary

Table 10: Qualitative summary of where post-quantum costs are expected to bind in Saudi and GCC sectors (derived from the analysis in this section; not measured).

Sector	Main constraint	Highest-priority action	Suitable techniques
Oil, gas and petrochemicals	SIS and DCS change control; remote SCADA links	PQ or hybrid firmware and logic signing in new DCS/SIS; procurement requirements	Hybrid ML-KEM on IP links; MACs for telemetry; gateways for remote segments
Power and inter-connection	Millisecond protection timing; serial field links; cross-border coordination	PQ or hybrid key establishment for existing MAC-based protection	Hybrid TLS for telecontrol; regional algorithm coordination
Desalination and water transmission	Low-bandwidth remote stations; legacy RTUs	Protect gateways and remote access; PQ-capable equipment in new projects	Hybrid VPNs; OTCC segmentation; replacement plans
Banking and payments	HSM support; international interoperability	Hybrid key establishment on external links; PKI readiness	Hybrid TLS; ML-DSA in PKI
Government services	Long-lived records; national PKI	Hybrid ML-KEM for data in transit; plan PQ or hybrid roots	Hybrid TLS; staged PKI migration

8 Mapping PQC Migration to Regional Frameworks

Saudi CI operators already demonstrate compliance with NCA and, where applicable, SAMA frameworks through regular assessments. Embedding PQC migration within these existing processes is more practical than creating parallel programmes. This section maps migration activities to the relevant framework areas and suggests evidence that cybersecurity assessors could request. The mapping is at the level of framework areas rather than individual control numbers, because control numbering differs between framework versions; practitioners should map to the specific controls of the versions in force, using the binding Arabic texts.

8.1 Activity-to-framework mapping

Table 11: Mapping of PQC migration activities to areas of NCA and SAMA frameworks (proposed; not an official interpretation).

Migration activity	Framework areas	Notes
Cryptographic inventory, including firmware keys and trust anchors	NCA ECC asset management and cryptography; OTCC asset management; SAMA CSF asset management	Extends existing asset registers with algorithm, key size, key location and device retirement date.
Quantum risk assessment	NCA ECC cybersecurity risk management; OTCC risk management; SAMA CSF risk management	Apply Mosca’s inequality per asset (Table 13).
Approved algorithms	NCA ECC, CCC and CCCC cryptography controls, which reference NCS [52]	Post-quantum algorithms become usable for compliance once included in the finalized NCS-2 [53].
Firmware and logic integrity	OTCC defence controls for ICS; secure configuration and change management	Post-quantum or hybrid signatures for firmware and engineering changes in new equipment.
Procurement requirements	NCA ECC and OTCC third-party cybersecurity; SAMA CSF third-party security	Require PQ-capable secure boot, crypto-agility and vendor PQC roadmaps.
Key establishment on external links	NCA ECC network security and cryptography; SAMA CSF cryptography	Hybrid ML-KEM where TLS or IPsec is available.
Resilience and fallback	NCA ECC and OTCC resilience domains	Test that PQ changes do not break failover, and that fallback cannot downgrade to classical-only.

8.2 Guidance for cybersecurity assessors

Assessors conducting NCA or SAMA assessments are well placed to identify post-quantum exposure early, because they already review cryptography, asset management and third-party controls. Without adding new compliance requirements, an assessment could collect the following evidence as observations to support future planning:

1. Whether the cryptographic inventory records public-key algorithms and key sizes, including those embedded in OT devices, firmware verification and PKI roots.
2. Which NCS level (MODERATE or ADVANCED) applies to each system, and whether the entity is tracking the NCS-2 update.
3. Whether procurement specifications for OT and network equipment ask vendors about post-quantum capability and crypto-agility.
4. Whether data classification records how long confidentiality must be maintained, which determines harvest-now-decrypt-later exposure.
5. Whether equipment with non-upgradable firmware trust anchors has recorded retirement dates.

These observations would give regulators an evidence base on sector readiness and give entities an early start on the inventory and prioritization phases of migration.

9 Migration Roadmap and Algorithm Selection

9.1 International timelines

Several governments have now published explicit milestones. Table 12 compares international and regional instruments, and Figure 5 places them on a common axis. Although their legal status and scope differ, the international milestones converge on two dates: around 2030 for high-priority or high-risk systems and 2035 for completion. Regional instruments have so far focused on approved algorithms (Saudi Arabia) and mandatory transition planning (UAE) rather than completion dates. Because Gulf operators buy most of their equipment and software from international vendors, whose product roadmaps follow U.S. and European deadlines, those deadlines will shape the region’s migration in practice even where no local date has been set.

Table 12: Published post-quantum migration milestones, international and regional.

Source	Milestones
United States: NSM-10 (2022) [14]; NIST IR 8547 (draft, 2024) [12]	Federal goal of mitigating quantum risk by 2035; quantum-vulnerable algorithms at 112-bit security deprecated after 2030, all disallowed after 2035 (proposed).
United States: CNSA 2.0 (2022) [13]	National security systems: stateful hash-based signatures for firmware signing; specified PQC algorithms (ML-KEM-1024, ML-DSA-87) with transition across product categories through 2035.
United Kingdom: NCSC (2025) [15]	By 2028: migration goals, full discovery and initial plan. By 2031: highest-priority migrations. By 2035: complete migration.
European Union: Recommendation (2024) [16] and NIS Cooperation Group roadmap (2025) [17]	By end 2026: national roadmaps, inventories and first steps. By end 2030: high-risk use cases migrated and quantum-safe software and firmware updates by default. By 2035: as many remaining systems as feasible.
Saudi Arabia: NCA NCS-1:2020 [52]; NCS-2:2025 draft [53]	Post-quantum cryptography addressed in an NCS-1 appendix (2020). Draft NCS-2 (public consultation Dec. 2025) adds post-quantum algorithm standards. No dated migration deadline published at the time of writing.
United Arab Emirates: National Encryption Policy (Nov. 2025) [58, 59]	Government entities must develop approved PQC transition plans; National Post-Quantum Migration Programme under the UAE Cybersecurity Council.

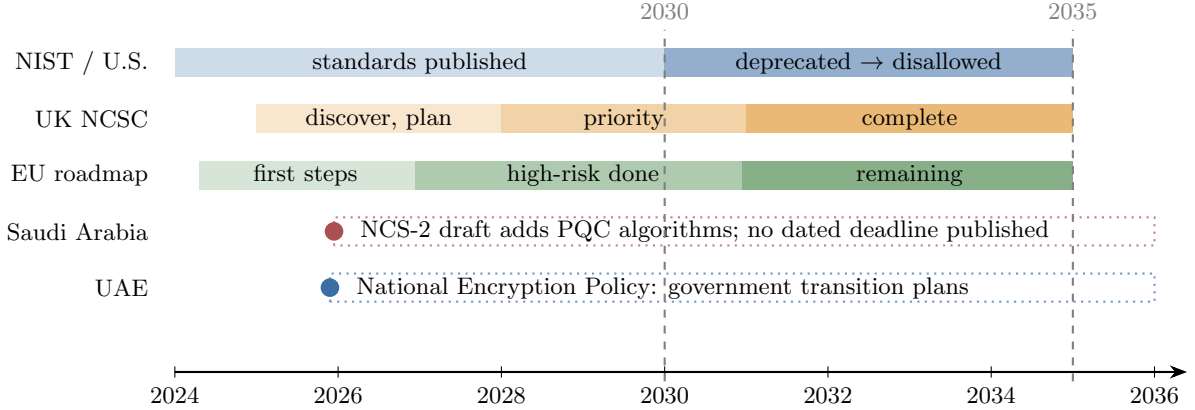


Figure 5: Published migration milestones on a common timeline (simplified from Table 12). Dots mark the Saudi NCS-2 consultation (Dec. 2025) and the UAE National Encryption Policy (Nov. 2025); NCS-1 has addressed PQC in an appendix since 2020.

For Gulf CI operators, the practical implication of these timelines combined with equipment lifetimes is that devices procured from now on will, in most cases, still be in service after 2035. Procurement is therefore the most important lever available today, and it does not depend on a local deadline being set.

9.2 Phased migration roadmap

Figure 6 proposes a four-phase migration roadmap for Saudi and GCC CI operators. In the absence of a regional completion deadline, its target dates follow the international milestones that will drive vendor roadmaps, and its activities are designed to fit within existing NCA OTCC and NCA ECC processes (Section 8). It is also consistent with the joint CISA, NSA and NIST guidance on quantum readiness [18], and it incorporates the findings of this paper: that bandwidth and trust anchors, rather than computation, are the main constraints. The roadmap is a proposal derived from the analysis above and has not been validated in operational settings.

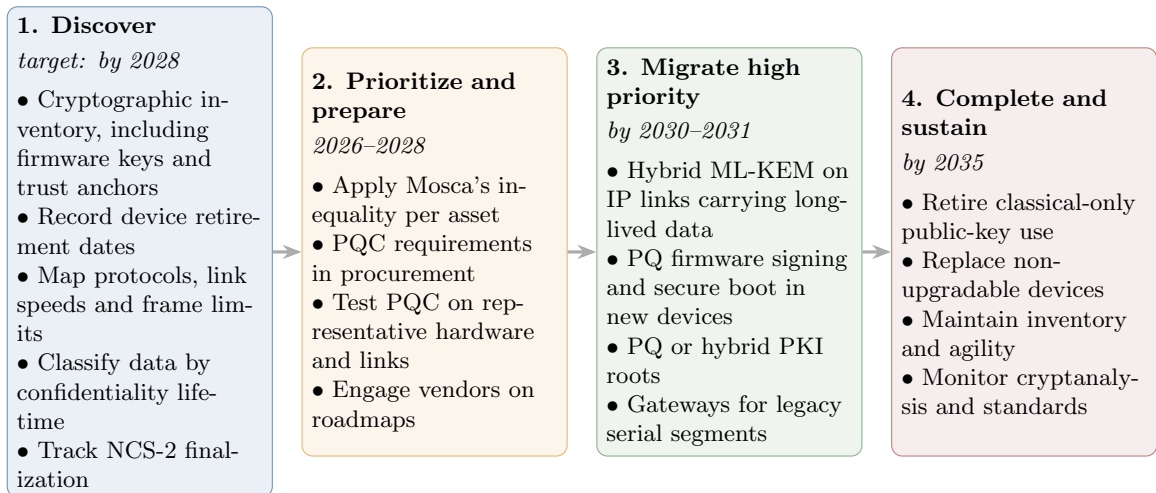


Figure 6: Proposed phased migration roadmap for Saudi and GCC CI operators, aligned with international milestones and regional frameworks.

9.3 Prioritizing assets with Mosca’s inequality

Table 13 applies Mosca’s inequality qualitatively to typical CI assets. For confidentiality, x is how long the data must remain secret; for authenticity, the relevant horizon is how long the device or trust anchor will remain in service. The values are illustrative ranges to show the reasoning, not measurements.

Table 13: Illustrative application of Mosca’s inequality to CI assets.

Asset	Threat type	Horizon (x or service life)	Implication
Real-time control commands	Authenticity	Seconds (message); device life for keys	Symmetric MACs suffice per message; key establishment must migrate before a CRQC
Operational telemetry	Confidentiality	Hours to months	Low harvest-now-decrypt-later exposure
Engineering drawings, network architecture	Confidentiality	Years to decades	High exposure; migrate key establishment early
Patient and financial records	Confidentiality	Decades	High exposure; migrate key establishment early
Firmware trust anchors in new devices	Authenticity	Device life, 15–20+ years	Must be post-quantum capable at procurement; cannot be fixed later
PKI root certificates	Authenticity	10–25 years	Plan PQ or hybrid roots early; long issuance lead times

9.4 Algorithm selection guide

Figure 7 condenses the benchmark and framing results into a selection guide. It reflects the standards and results available at the time of writing and should be revisited as FN-DSA and HQC are finalized and as embedded benchmarks become available. Entities subject to NCA frameworks must in addition use algorithms and parameter sets approved in the version of NCS in force at the time of deployment.

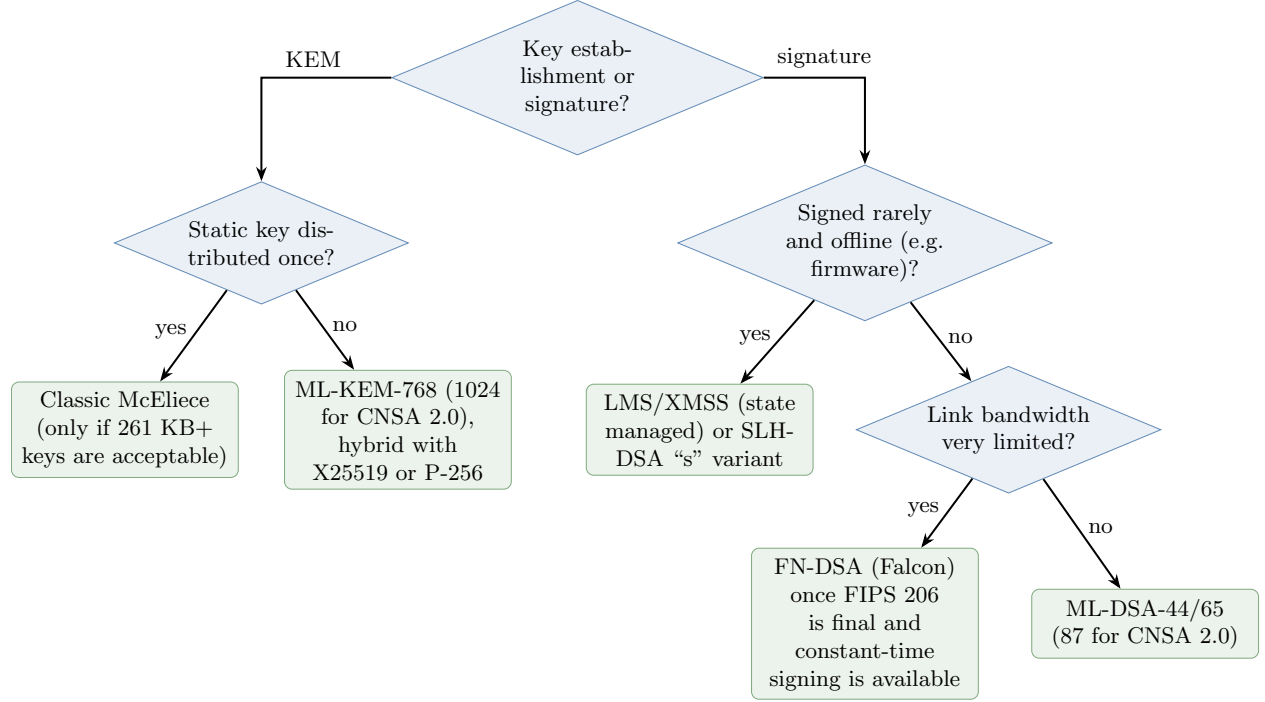


Figure 7: Algorithm selection guide derived from the results. For per-message protection on very constrained or time-critical links, prefer symmetric MACs keyed through a post-quantum key exchange.

10 Discussion

10.1 Computation is not the main barrier on capable hardware

Earlier discussions of PQC in CI often assume that post-quantum algorithms are uniformly much slower than classical ones. On the processor measured here that is not the case for the primary standards: ML-KEM outperformed ECDH, ML-DSA verification matched or beat ECDSA, and ML-DSA signing was only four to nine times slower than ECDSA and faster than RSA. These results agree in direction with earlier TLS-level studies on server hardware, which found that lattice KEMs add little handshake latency and that signature size, not computation, dominates the cost of post-quantum authentication [38, 39].

This finding does not transfer automatically to embedded controllers. Microcontroller benchmarks such as the pqm4 project show that the same algorithms consume far more cycles and, critically, more stack and RAM on ARM Cortex-M4 devices [40], and many PLCs and remote terminal units are older or weaker still. Whether PQC is computationally feasible on a given field device remains an empirical question that this study does not answer.

10.2 Size, framing and bandwidth are the dominant costs

The framing analysis shows where legacy CI links are likely to break. Serial links running at 9,600 to 19,200 bit/s remain common in utilities and water systems, and on them the transmission time of a single PQC signature is measured in seconds. The practical consequences are that per-message post-quantum signatures are unrealistic on such links; public-key operations should be confined to infrequent tasks such as key establishment, key updates and firmware updates; and frequent messages should continue to be protected by symmetric MACs keyed through a post-quantum key exchange. The design of DNP3 Secure Authentication, which already uses HMACs for routine messages [34], fits this model; what must change is how its keys are established and updated.

10.3 The urgency is different for confidentiality and for authenticity

For confidentiality, the harvest-now-decrypt-later threat applies today to any traffic whose value outlasts the arrival of a CRQC. In CI, operational telemetry typically loses its value quickly, but engineering data, network architecture documentation, patient records and financial records may need protection for years or decades. Key establishment protecting such data should be migrated first, and hybrid ML-KEM makes this feasible wherever TLS or similar protocols are already used.

For authenticity, recorded signatures cannot be forged retroactively, so the threat begins only when a CRQC exists. In CI, however, the relevant question is how long equipment deployed now will remain in service. A controller commissioned in 2026 with a 20-year life will still be operating in 2046, more than a decade after the proposed 2035 disallowance. If its boot-time firmware verification key is ECDSA or RSA only, and stored in one-time-programmable memory, it cannot be migrated without replacing hardware. The Triton attack, which targeted the logic of safety controllers at a Saudi petrochemical facility [48], shows why the authenticity of controller code matters so much in this region. For this reason, firmware trust anchors in newly procured devices are arguably the most urgent CI migration item, even though they are not exposed to harvest-now-decrypt-later attacks. The EU roadmap makes the same point by asking for quantum-safe software and firmware updates by default by 2030 [17]. Firmware signing is also well matched to PQC: signing happens rarely and offline, where the cost of SLH-DSA or stateful LMS/XMSS is acceptable, and verification at boot

is fast. This is the use case CNSA 2.0 assigns to stateful hash-based signatures [13, 24].

10.4 Regional considerations

Three features of the Gulf context deserve emphasis. First, operators depend heavily on international vendors and engineering contractors for OT equipment, so the region’s migration pace will largely follow vendor roadmaps shaped by U.S. and European deadlines. Regional regulators can influence this most effectively through procurement requirements and through the approved-algorithm list in NCS, rather than by developing separate algorithms. Second, Vision 2030 emphasizes local content and the localization of technology, and the Aramco and Pasqal partnership explicitly includes training and joint research [51]; a comparable investment in local expertise in post-quantum implementation, testing and assessment would reduce dependence on external suppliers during the transition. Third, shared infrastructure such as the GCC grid interconnection and cross-border financial links means that some migrations can only succeed if neighbouring states choose compatible algorithms at compatible times, which calls for regional coordination.

10.5 Implementation risk

New implementations carry new risks. Side-channel and fault attacks on lattice schemes are an active research area [41], and implementation bugs have appeared even in widely reviewed code: the KyberSlash timing vulnerabilities affected several Kyber implementations through secret-dependent division [42]. Field devices are often physically accessible to attackers, which makes power and electromagnetic side channels more relevant than in data centres. Falcon’s floating-point Gaussian sampling is particularly difficult to implement in constant time on devices without suitable floating-point hardware. CI operators should prefer validated implementations and ask vendors how side-channel resistance was assessed.

Hybrid schemes add their own risks: two algorithms mean more code, larger messages and more complex negotiation, and poorly designed fallback can allow downgrade to classical-only operation. Hybrids should therefore be treated as a transition mechanism with a defined end state, not as permanent architecture.

10.6 Economic considerations

This study did not measure costs, but its results indicate where they are likely to concentrate. Replacing computation-bound components is unlikely to be the main expense on modern hardware; the larger costs are hardware replacement for devices without upgradeable trust anchors or firmware, bandwidth upgrades or gateway deployments for serial and radio links, re-engineering of protocols and certificate infrastructure for larger messages, and testing and recertification in regulated and safety-critical environments. Aligning migration with planned equipment refresh cycles, and adding PQC requirements to procurement now, avoids paying twice for the same assets.

10.7 Comparison with common assumptions

Table 14 contrasts several assumptions that are common in discussions of PQC in CI with the evidence in this paper.

Table 14: Common assumptions compared with the evidence from this study.

Assumption	Evidence from this study
PQC is uniformly much slower than ECC	On an M2 Pro, ML-KEM is faster than ECDH and ML-DSA verification is as fast as or faster than ECDSA (Tables 5 and 6).
Hash-based signatures are slow to verify	SLH-DSA verification is $8\text{--}24\times$ ECDSA; signing is the bottleneck, at $1,200\text{--}25,000\times$ (Table 7).
Key size is a minor issue	On a 9,600 bit/s DNP3 link one ML-DSA-44 signature takes about 3 s to transmit (Table 9).
Authentication can wait until a CRQC is near	Firmware trust anchors in devices bought today will still be in use after 2035 and cannot be changed later.
All promising candidates are safe to plan for	Rainbow and SIKE were broken late in standardization [25, 26].

11 Recommendations

11.1 For critical infrastructure operators in Saudi Arabia and the GCC

1. Extend existing NCA and SAMA asset and cryptography inventories to record public-key algorithms, key sizes, firmware-signing keys, embedded trust anchors and expected device retirement dates.
2. Classify data by required confidentiality lifetime and migrate key establishment for long-lived data first, using hybrid ML-KEM where TLS or IPsec is available, subject to the algorithms approved in the NCS version in force.
3. Require post-quantum-capable secure boot and firmware verification, and crypto-agility, in procurement of OT and network equipment expected to operate past 2030, with particular priority for DCS and safety instrumented systems.
4. Keep frequent, time-critical messages on symmetric MACs, and move key establishment and key updates to post-quantum or hybrid mechanisms.
5. For remote pipeline, pumping and field sites with legacy serial or radio links, use gateways at the boundary of the legacy segment and apply OTCC segmentation as a compensating control.
6. Test PQC on representative hardware and links before deployment, including worst-case timing and multi-fragment behaviour.
7. Assign ownership for cryptographic migration within the cybersecurity function required by NCA frameworks, and build internal expertise.

11.2 For regulators (NCA, SAMA and GCC counterparts)

1. Finalize and publish NCS-2 with approved post-quantum algorithms and parameter sets, and clear guidance on hybrid use during the transition.
2. Consider publishing indicative migration milestones aligned with international dates (around 2030 for high-risk systems and 2035 for completion), so that operators and vendors can plan investment.
3. Incorporate cryptographic inventory, quantum risk assessment and PQ-capable procurement into OTCC and NCA ECC guidance and assessment tools.
4. Coordinate algorithm choices and timing for cross-border systems, such as the GCC grid interconnection and financial links, through regional bodies.
5. Support smaller operators with shared services, guidance and testing facilities.

11.3 For vendors and integrators serving the region

1. Ship post-quantum-capable firmware verification in new products, with field-updatable algorithms where possible, and document support for NCS-approved algorithms.
2. Publish memory and timing figures for PQC on the controllers actually shipped, not only on server hardware.

3. Separate cryptographic functions into replaceable modules with stable interfaces, and use validated, constant-time implementations.
4. Build local implementation and support capability in line with Vision 2030 localization goals.

11.4 For cybersecurity assessors and GRC practitioners

1. Record post-quantum readiness observations during NCA and SAMA assessments, using the evidence items in Section 8.
2. Treat embedded trust anchors and firmware-signing keys as cryptographic assets within scope of inventory reviews.
3. Review procurement specifications for crypto-agility and post-quantum capability as part of third-party cybersecurity assessments.

11.5 For standards bodies and researchers

1. Add algorithm negotiation and support for fragmented public-key material to industrial security profiles, and post-quantum or hybrid options for key establishment in DNP3 Secure Authentication, IEC 62351 and Modbus/TCP Security.
2. Develop compact certificate profiles for bandwidth-constrained environments.
3. Benchmark PQC on the controller families most widely deployed in Gulf infrastructure, and publish the results.

12 Limitations

Single platform. The benchmarks were run once, on a single desktop-class processor. Absolute times will differ on other hardware, and the relative costs may differ substantially on microcontrollers, where memory rather than time is often the binding constraint.

SLH-DSA data provenance. The SLH-DSA 192- and 256-bit parameter sets were measured only in the initial full-suite run. The confirmation run of the 128-bit sets agreed with it to within 1.3%, but the larger sets were not repeated.

Unequal optimization. OpenSSL includes hand-tuned assembly for P-256 and RSA on ARM64, and the extent to which each liboqs algorithm used architecture-specific code paths was not independently verified. The comparison is therefore between typical library implementations rather than between algorithms in the abstract.

Measurements not taken. Energy consumption, memory usage and end-to-end protocol latency were not measured.

Theoretical framing analysis. The framing results are lower bounds computed from protocol specifications. Real deployments add application headers, retransmissions and link turnaround.

Regional sources. The regional analysis relies on public English-language versions of Saudi and UAE instruments and on press reports for the most recent developments. NCS-2 was analysed as a consultation draft, and its final content may differ. Framework control numbering was deliberately not used, because it differs between versions.

No operational data. The sector analysis relies on standards, regulation and published literature rather than on data from operational systems in the region, and the migration roadmap and framework mapping have not been validated with operators or regulators.

13 Conclusion and Future Work

Saudi Arabia and the GCC combine three conditions that make post-quantum migration both urgent and difficult: economies built on long-lived operational technology, a documented history of sophisticated attacks on industrial and safety systems, and critical services that depend on dispersed, often low-bandwidth control networks. The region’s regulators have begun to respond, with PQC in the NCA’s National Cryptographic Standards since 2020, a draft update adding post-quantum algorithm standards, and a mandatory planning requirement for UAE government entities.

This paper’s measurements show that standardized PQC is computationally modest on capable processors: ML-KEM and ML-DSA performed comparably to, and in several operations better than, the classical algorithms they replace. The serious obstacles lie elsewhere. PQC signatures and key-exchange messages are roughly ten to several hundred times larger than their elliptic-curve counterparts (Classic McEliece public keys, thousands of times), which on legacy serial and radio links turns a single signature into seconds of transmission; and devices bought today may carry classical-only firmware trust anchors well past 2035. For the Gulf, the most effective near-term actions are therefore to add post-quantum capability to procurement, to protect firmware and controller logic with post-quantum or hybrid signatures in new equipment, to keep frequent messages on symmetric protection while migrating key establishment, and to embed these steps in the existing NCA and SAMA compliance processes that operators already follow.

13.1 Future work

1. **Embedded benchmarks.** Repeat the measurements on an ARM Cortex-M4 development board using pqm4 and on a Raspberry Pi as a proxy for gateway-class hardware, including stack, RAM and energy.
2. **Protocol measurements.** Measure end-to-end handshakes over emulated low-bandwidth links using the OpenSSL OQS provider.
3. **Industrial testbed.** Build a DNP3 or Modbus testbed to measure multi-fragment behaviour and the effect of PQC key updates on polling cycles.
4. **Regional readiness study.** With appropriate ethical approval and permission, survey Saudi and GCC operators on cryptographic inventory and PQC readiness.
5. **NCS-2 analysis.** Analyse the final NCS-2 once published, including its approved post-quantum algorithms and their fit with the constraints identified here.

Data availability

The raw benchmark outputs, parsing and analysis scripts, and figure-generation code are included with this paper’s source files (`data/`).

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A Reproducing the Benchmarks

All benchmarks were run on an Apple M2 Pro with macOS 26.5.1. The software used was Apple clang 21, OpenSSL 3.6.4 from Homebrew, and liboqs 0.16.0 [43] at commit b196b57. The full commit hash is recorded in `data/setup.txt`.

```
brew install cmake ninja openssl@3
git clone --depth 1 https://github.com/open-quantum-safe/liboqs.git
cd liboqs && mkdir build && cd build
cmake -GNinja -DOQS_USE_OPENSSL=ON -DOPENSSL_ROOT_DIR=$(brew --prefix openssl@3) ..
ninja

for a in ML-KEM-512 ML-KEM-768 ML-KEM-1024 Classic-McEliece-348864; do
  ./tests/speed_kem $a; done > kem.txt
for a in ML-DSA-44 ML-DSA-65 ML-DSA-87 Falcon-512 Falcon-1024 \
  SLH_DSA_PURE_SHA2_128F SLH_DSA_PURE_SHA2_128S; do
  ./tests/speed_sig $a; done > sig.txt

$(brew --prefix openssl@3)/bin/openssl speed ecdsap256 ecdhp256 rsa2048
```

The `speed_kem` and `speed_sig` tools accept one algorithm name per invocation, which is why the commands above loop over algorithms. The files `data/parse.py` and `data/analysis.py` regenerate every table value and figure in this paper from the raw outputs.

B DNP3 Framing: Worked Example

Consider an ML-DSA-44 signature of $P = 2,420$ octets sent over a DNP3 serial link.

1. The transport layer splits the payload into segments of at most 249 octets: nine full segments ($9 \times 249 = 2,241$ octets) and one segment of 179 octets, for 10 frames.
2. Each segment gains a 1-octet transport header, so the frames carry $n = 250$ user-data octets (nine times) and $n = 180$ (once).
3. A full frame occupies $L(250) = 10 + 250 + 2\lceil 250/16 \rceil = 10 + 250 + 32 = 292$ octets; the last occupies $L(180) = 10 + 180 + 2 \times 12 = 214$ octets.
4. The total is $9 \times 292 + 214 = 2,842$ octets, or 28,420 bits at 10 bits per octet.
5. At 9,600 bit/s this takes $28,420/9,600 \approx 2.96$ s; at 19,200 bit/s, about 1.48 s.

For comparison, a 64-octet ECDSA signature fits in one frame with $n = 65$, occupying $L(65) = 10 + 65 + 2 \times 5 = 85$ octets, or about 0.09 s at 9,600 bit/s.

C Glossary

CRQC

Cryptographically relevant quantum computer: a quantum computer able to break deployed public-key cryptography.

CNI Critical national infrastructure.

Cryptographic agility

The ability to replace cryptographic algorithms without redesigning the system.

DNP3

Distributed Network Protocol 3 (IEEE 1815), widely used in electric and water utilities.

DCS Distributed control system: the main process control system of a plant.

ECC (NCA)

Essential Cybersecurity Controls issued by Saudi Arabia's National Cybersecurity Authority (not to be confused with elliptic-curve cryptography).

FN-DSA

The NIST name for Falcon, being standardized as FIPS 206.

GCC Gulf Cooperation Council: Bahrain, Kuwait, Oman, Qatar, Saudi Arabia and the United Arab Emirates.

GOOSE

Generic Object Oriented Substation Event, a time-critical multicast message type in IEC 61850.

Harvest now, decrypt later

Recording encrypted traffic today in order to decrypt it once a CRQC exists.

Hybrid scheme

A combination of a classical and a post-quantum algorithm that remains secure if either is unbroken.

KEM

Key-encapsulation mechanism: a public-key method for establishing a shared secret key.

LMS / XMSS

Stateful hash-based signature schemes standardized in NIST SP 800-208.

ML-DSA

Module-Lattice-Based Digital Signature Algorithm (FIPS 204), formerly CRYSTALS-Dilithium.

ML-KEM

Module-Lattice-Based Key-Encapsulation Mechanism (FIPS 203), formerly CRYSTALS-Kyber.

NCA National Cybersecurity Authority of Saudi Arabia.

NCS National Cryptographic Standards issued by the NCA (NCS-1:2020; NCS-2:2025 draft).

OT Operational technology: systems that monitor and control physical processes.

OTCC

Operational Technology Cybersecurity Controls issued by the NCA (OTCC-1:2022).

PLC / RTU

Programmable logic controller / remote terminal unit: field controllers in industrial systems.

SAMA

Saudi Central Bank, which issues the SAMA Cyber Security Framework.

SIS Safety instrumented system: an independent controller that brings a process to a safe state when dangerous conditions arise.

SLH-DSA

Stateless Hash-Based Digital Signature Algorithm (FIPS 205), formerly SPHINCS+.

Trust anchor

A public key built into a device or system and used to verify firmware, certificates or other signatures.